Администрирование локальных сетей

Использование протокола STP. Агрегирование каналов (Лабораторная работа №9)

Суннатилло Махмудов

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Цели и задачи работы

Цель лабораторной работы

- Изучение возможностей протокола **STP**
- Анализ работы **Rapid PVST+**
- Исследование отказоустойчивости сети при наличии резервных каналов
- Знакомство с режимом **PortFast**
- Изучение принципов агрегирования каналов **EtherChannel**

Теоретическая часть

Назначение STP

- **STP (Spanning Tree Protocol)** предотвращает образование петель в сети второго уровня
- При наличии избыточных соединений отключает лишние каналы логически
- Обеспечивает корректную работу сети при сохранении резервных путей

Выполнение лабораторной работы

Формирование резервного соединения

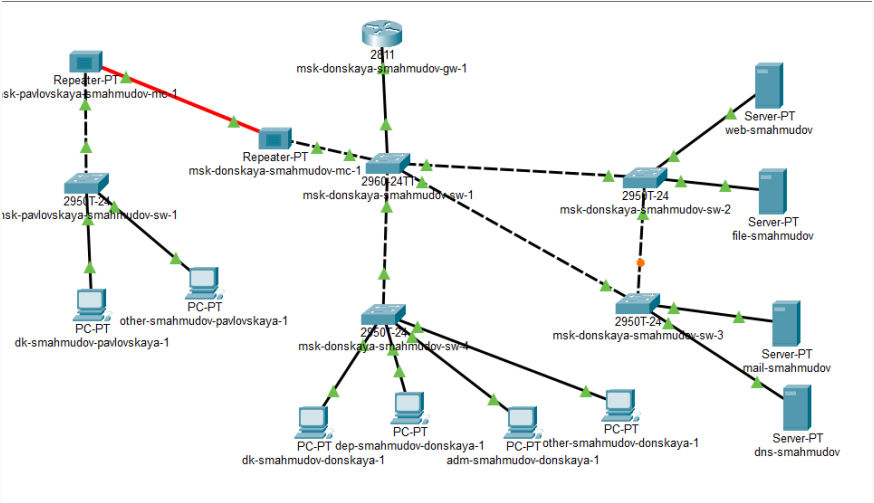

Рис. 1: Топология сети после формирования резервного соединения

Проверка прохождения ICMP-пакетов

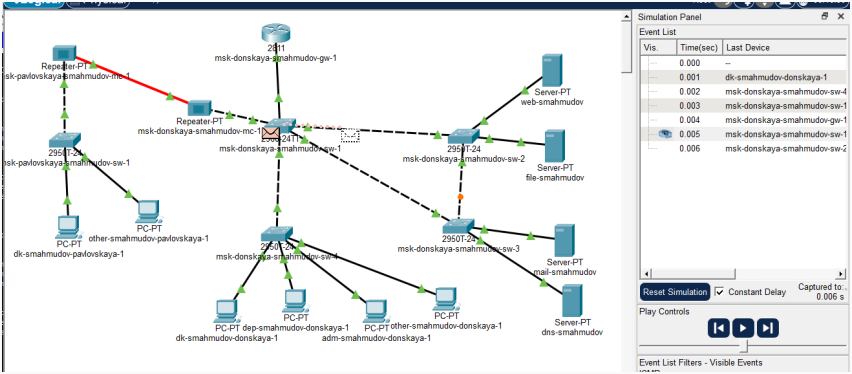

Рис. 2: Передача ICMP-пакетов в режиме Simulation

Маршрут ICMP через коммутатор sw-2

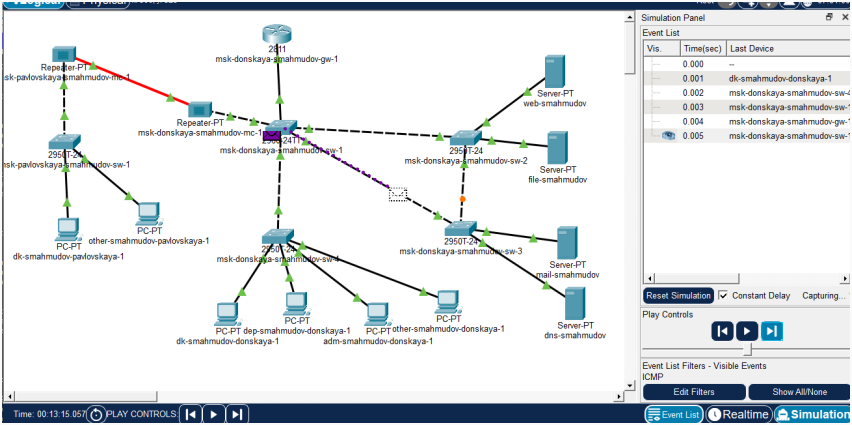

Рис. 3: Прохождение ICMP-пакета через коммутатор msk-donskaya-smahmudov-sw-2

Анализ STP для VLAN 3 на sw-2

```
Password:
msk-donskaya-smahmudov-sw-2#show span
msk-donskaya-smahmudov-sw-2#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0002.4A9D.A4D1
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32771  (priority 32768 sys-id-ext 3)
             Address     000A.41D5.5C7E
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/2            Desg FWD 4         128.26   P2p
Gi0/1            Root FWD 4         128.25   P2p

msk-donskaya-smahmudov-sw-2#
```

Рис. 4: Просмотр состояния STP на коммутаторе msk-donskaya-smahmudov-sw-2

Назначение корневого коммутатора

```
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-smahmudov-sw-1(config)#spa
msk-donskaya-smahmudov-sw-1(config)#spanning-tree root pri
msk-donskaya-smahmudov-sw-1(config)#spanning-tree vlan 3 ro
msk-donskaya-smahmudov-sw-1(config)#spanning-tree vlan 3 root pr
msk-donskaya-smahmudov-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-smahmudov-sw-1(config)#exit
msk-donskaya-smahmudov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-smahmudov-sw-1#show spa
msk-donskaya-smahmudov-sw-1#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    24579
             Address     0002.4A9D.A4D1
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24579  (priority 24576 sys-id-ext 3)
             Address     0002.4A9D.A4D1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/24           Desg FWD 19        128.24   P2p
Gi0/1            Desg FWD 4         128.25   P2p
Fa0/23           Desg FWD 19        128.23   P2p
Gi0/2            Desg FWD 4         128.26   P2p
Fa0/1            Desg FWD 19        128.1    P2p

msk-donskaya-smahmudov-sw-1#
```


Рис. 5: Настройка корневого коммутатора STP на msk-donskaya-smahmudov-sw-1

Изменение маршрутов после перестроения STP

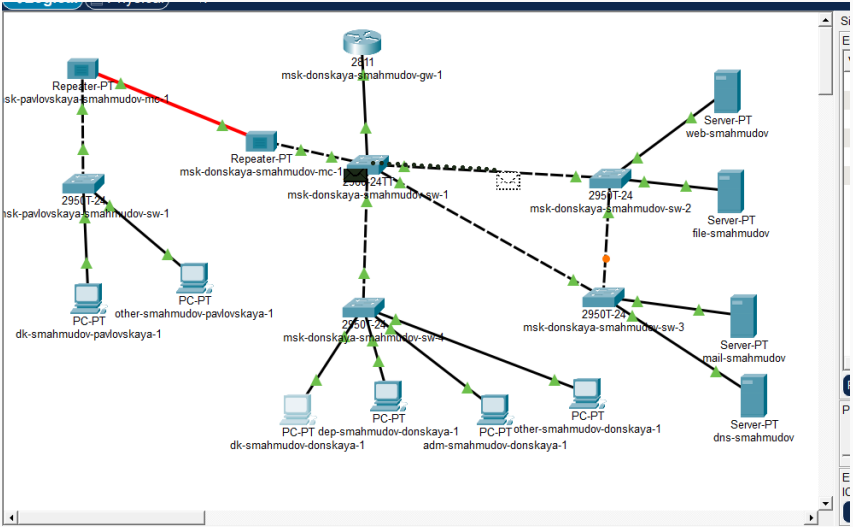

Рис. 6: Передача пакетов после перестроения дерева STP

Маршрут по резервному соединению

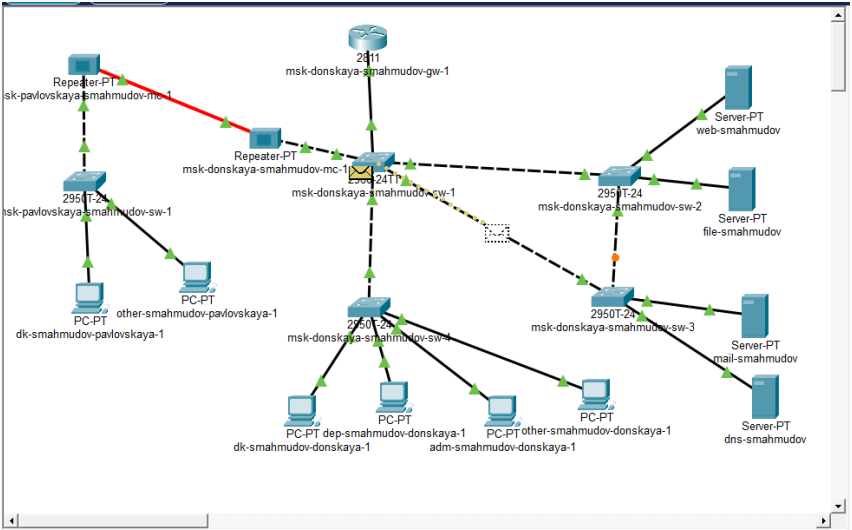

Рис. 7: Маршрут ICMP-пакетов по резервному соединению

Настройка PortFast

```
msk-donskaya-smahmudov-sw-2#
msk-donskaya-smahmudov-sw-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-smahmudov-sw-2(config)#int f0/1
msk-donskaya-smahmudov-sw-2(config-if)#spa
msk-donskaya-smahmudov-sw-2(config-if)#spanning-tree port
msk-donskaya-smahmudov-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-smahmudov-sw-2(config-if)#
msk-donskaya-smahmudov-sw-2(config-if)#int f0/2
msk-donskaya-smahmudov-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-smahmudov-sw-2(config-if)#
```

Рис. 8: Настройка режима PortFast на интерфейсах, подключённых к серверам

Отказоустойчивость STP

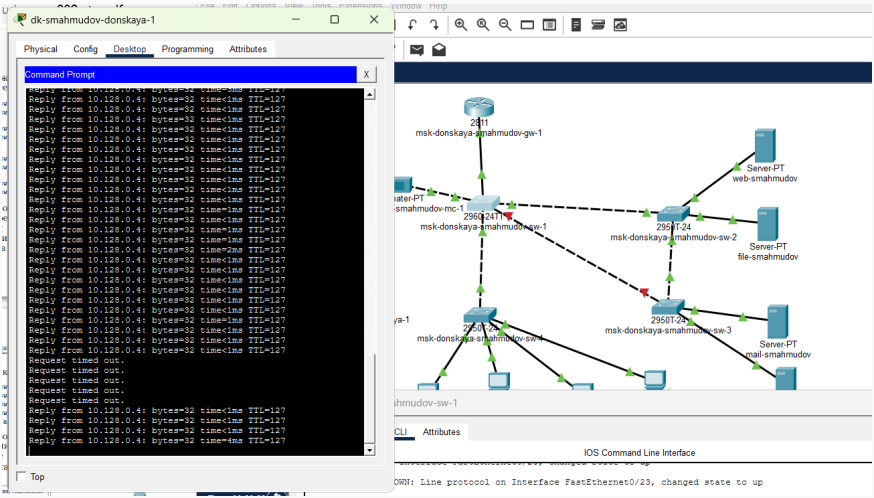

Рис. 9: Проверка отказоустойчивости STP при разрыве основного соединения

```
msk-donskaya-smahmudov-sw-1(config-if)#exit
msk-donskaya-smahmudov-sw-1(config)#
msk-donskaya-smahmudov-sw-1(config)#span mode rapi
msk-donskaya-smahmudov-sw-1(config)#span mode rapid-pvst
msk-donskaya-smahmudov-sw-1(config)#exit
msk-donskaya-smahmudov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-smahmudov-sw-1#wr mem
Building configuration...
[OK]
msk-donskaya-smahmudov-sw-1#
```

Рис. 10: Перевод коммутаторов в режим Rapid PVST+

Отказоустойчивость Rapid PVST+

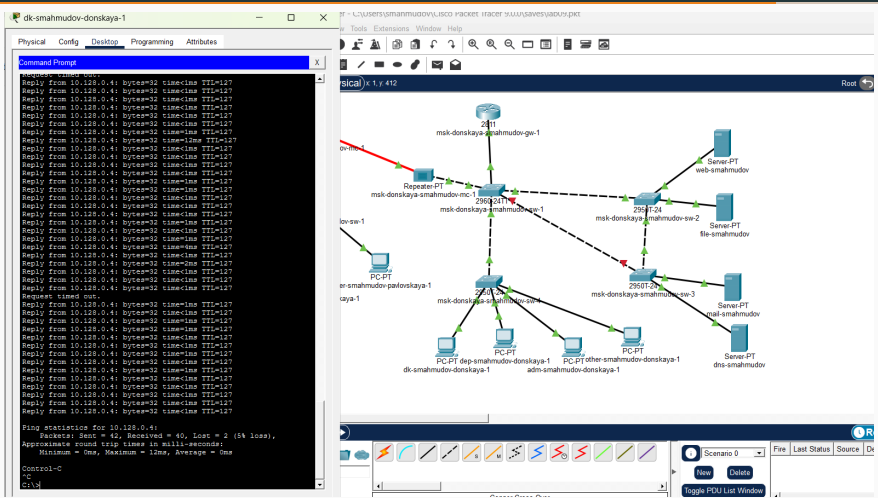

Рис. 11: Проверка отказоустойчивости Rapid PVST+

Формирование EtherChannel

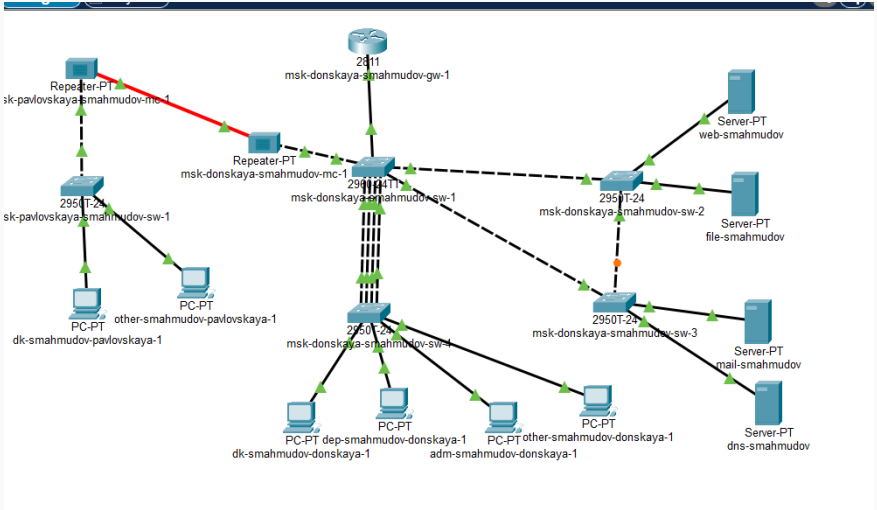

Рис. 12: Формирование агрегированного соединения между коммутаторами

Настройка EtherChannel на sw-1

```
msk-donskaya-smahmudov-sw-1(config-if)#exit
msk-donskaya-smahmudov-sw-1(config)#
msk-donskaya-smahmudov-sw-1(config)#int range f0/20 - f0/23
msk-donskaya-smahmudov-sw-1(config-if-range)#cha
msk-donskaya-smahmudov-sw-1(config-if-range)#chann
msk-donskaya-smahmudov-sw-1(config-if-range)#channel-group 1 mode on
msk-donskaya-smahmudov-sw-1(config-if-range)#
Creating a port-channel interface Port-channel 1

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of Fa0/23 is full, Fa0/20 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of Fa0/23 is full, Fa0/21 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of Fa0/23 is full, Fa0/22 is
half)

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

msk-donskaya-smahmudov-sw-1(config-if-range)#exit
msk-donskaya-smahmudov-sw-1(config)#int portt-channel 1
                                       ^
% Invalid input detected at '^' marker.

msk-donskaya-smahmudov-sw-1(config)#int port-channel 1
msk-donskaya-smahmudov-sw-1(config-if)#sw mode trunk
msk-donskaya-smahmudov-sw-1(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/20 is not compatible with Fa0/23 and will be suspended (duplex of Fa0/20 is half, Fa0/23 is
full)

%EC-5-CANNOT_BUNDLE2: Fa0/21 is not compatible with Fa0/23 and will be suspended (duplex of Fa0/21 is half, Fa0/23 is
full)

%EC-5-CANNOT_BUNDLE2: Fa0/22 is not compatible with Fa0/23 and will be suspended (duplex of Fa0/22 is half, Fa0/23 is
full)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of Fa0/23 is full, Fa0/20 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of Fa0/23 is full, Fa0/21 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of Fa0/23 is full, Fa0/22 is
half)

msk-donskaya-smahmudov-sw-1(config-if)#
```

Copy Paste

Настройка EtherChannel на sw-4

```
msk-donskaya-smahmudow-sw-4#
msk-donskaya-smahmudow-sw-4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-smahmudow-sw-4(config)#int range f0/20 - 23
msk-donskaya-smahmudow-sw-4(config-if-range)#no sw access vlan 104
msk-donskaya-smahmudow-sw-4(config-if-range)#exit
msk-donskaya-smahmudow-sw-4(config)#
msk-donskaya-smahmudow-sw-4(config)#int range f0/20 - 23
msk-donskaya-smahmudow-sw-4(config-if-range)#channel-group 1 mode on
msk-donskaya-smahmudow-sw-4(config-if-range)#
Creating a port-channel interface Port-channel 1

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of Fa0/23 is full, Fa0/20 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of Fa0/23 is full, Fa0/21 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of Fa0/23 is full, Fa0/22 is
half)

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to down

msk-donskaya-smahmudow-sw-4(config-if-range)#exit
msk-donskaya-smahmudow-sw-4(config)#int port-channel 1
msk-donskaya-smahmudow-sw-4(config-if)#sw mode trunk
msk-donskaya-smahmudow-sw-4(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/20 is not compatible with Fa0/23 and will be suspended (duplex of Fa0/20 is half, Fa0/23 is
full)

%EC-5-CANNOT_BUNDLE2: Fa0/21 is not compatible with Fa0/23 and will be suspended (duplex of Fa0/21 is half, Fa0/23 is
full)

%EC-5-CANNOT_BUNDLE2: Fa0/22 is not compatible with Fa0/23 and will be suspended (duplex of Fa0/22 is half, Fa0/23 is
full)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of Fa0/23 is full, Fa0/20 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of Fa0/23 is full, Fa0/21 is
half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of Fa0/23 is full, Fa0/22 is
half)

msk-donskaya-smahmudow-sw-4(config-if)#exit
```

Итоги работы

Выводы

- Изучены принципы работы **STP** и **Rapid PVST+**
- Исследована отказоустойчивость сети при использовании резервных соединений
- Выполнена настройка **PortFast**
- Рассмотрен принцип агрегирования интерфейсов в **EtherChannel**
- Выявлено, что для корректной работы EtherChannel все объединяемые порты должны иметь одинаковые параметры